Advancing the GEOtiled Framework Through Scalable Terrain Parameter Computation

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ABSTRACT

The GEOtiled framework facilitates the scalable and efficient computation of high-resolution terrain parameters using Digital Elevation Models (DEMs) across the Continental United States (CONUS). These parameters are essential in Earth Science applications. This paper presents significant advances to optimizing GEOtiled's performance. In GEOtiled 2.0, we introduce three key optimizations to speed up GEOtiled's overall runtime: concurrent cropping, efficient mosaic operation, and unified parallel processing. These optimizations improve data distribution and handling, reducing computation times while maintaining accuracy. We evaluate performance using DEM data at 30-meter resolution over a region covering the state of Tennessee. Our results demonstrate that the optimized framework provides substantial performance improvements for generating terrain parameters. GEOtiled 2.0 software, openly available on GitHub, advances reproducible, data-driven scientific discovery in Earth Science.

KEYWORDS

High Performance Computing, Data Partitioning, Parallel Computing, Earth Science, Terrain Parameters

1 INTRODUCTION

Scientists use terrain parameters (e.g., elevation, slope, aspect) in their analysis of various Earth Science studies, such as predictive upsampling of soil moisture [4], forest fire propagation zoning [5], and ML-based hydrological mapping [2]. These parameters are derived from Digital Elevation Models (DEMs), some of which are available at high resolution from the United States Geological Survey (USGS) website. Various open-source Global Information System (GIS) libraries implement terrain parameters algorithms, and two commonly used ones are the Geospatial Data Abstraction Library (GDAL) [6] and the System for Automated Geoscientific Analyses (SAGA) [1]. However, neither of these libraries scales well as the data gets larger, causing slow execution times for processing larger DEMs. For example, the size of DEMs for the Continental United States (CONUS) at 10-meter resolution exceeds 300 GB. Therefore, a scalable method for computing terrain parameters is needed to meet the computational demands of these large, high-quality input datasets. We solve this issue by updating GEOtiled [3]: a workflow that integrates GDAL and SAGA libraries to provide rapid, scalable computation of terrain parameters by leveraging data decomposition and parallel computation. Our optimizations include: parallelizing the crop procedure for scalable computation, updating the mosaic algorithm to a better performing one, and unifying parallel processes in the workflow to minimize overhead.

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2 GEOTILED WORKFLOW

Before detailing our optimizations, we should first explain GEOtiled's 5-step process for providing parallel computation of terrain parameters (see Figure 1). In Step 1, GEOtiled downloads USGS's DEMs for a user-specified region and resolution. In Step 2, the downloaded DEMs are merged together and reprojected into a computable, preprocessed format (i.e., the file's units are converted from decimal degrees to meters). In Step 3, the preprocessed DEM is cropped into evenly-sized tiles. Because terrain parameter algorithms require knowledge of neighboring pixels to compute values at any given pixel, we preserve accuracy during computation by adding border pixels from neighboring tiles to each tile (i.e., a buffer). In Step 4, terrain parameters are computed in parallel for each cropped tile. In Step 5, computed tiles for each terrain parameter go through a mosaic process to produce a single file. These steps provide scientists a way to quickly generate and apply high spatial resolution terrain parameter data for their Earth Science applications.

A recent improvement from GEOtiled 1.0 to GEOtiled 2.0 is the addition of terrain parameter functions from the SAGA library, integrated into Step 4 of the workflow. GEOtiled 1.0 initially only used functions from GDAL to compute terrain parameters, but the SAGA library quintuples this number. However, functions from SAGA have higher computational costs than GDAL, and this cost is our primary motivation for introducing optimizations to GEOtiled 2.0.

3 OPTIMIZATIONS

3.1 Optimization 1: Concurrent Cropping

For our first optimization, we parallelize Step 3 of the GEOtiled workflow: the crop process. Originally, the crop procedure computed each bound, or window, based on a specified tile size and then cropped the tile from the input DEM. This process is repeated until all tiles are cropped. Our new implementation computes all

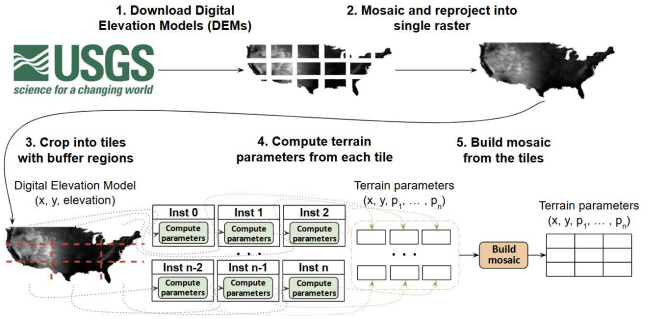


Figure 1: GEOtiled workflow and its drive steps (i.e., download Digital Elevation Models, mosaic and reproject into single raster, crop into tiles with buffer regions, compute terrain parameters for each time, and build a mosaic from the tiles).

the windows first and then crops all tiles in parallel for the given list of windows. This provides us with scalable performance as we continue to reduce our tile size and subsequently produce more tiles.

3.2 Optimization 2: Updated Mosaic Algorithm

Our second optimization reworks Step 5, the mosaic procedure, in the GEOtiled workflow. The previous method averages the overlapping pixels in each tile's shared region. This process does not scale well because more shared areas must be averaged together with an increasing number of tiles to mosaic. Instead, our optimization crops off the buffer regions and concatenates tiles into a single file. We can opt to crop the buffer instead of averaging because the pixels added during Step 3, the crop step, are copied from the borders of neighboring tiles and not a shared region of pixels between tiles. We also parallelize the cropping of buffer regions across tiles to provide scalable performance.

3.3 Optimization 3: Unified Parallel Processes

Our final optimization of the GEOtiled workflow is the unification of the multiple parallel processes, some of which are newly introduced by our two previous optimizations. The concurrent cropping of the input DEM, parallel computation of terrain parameters, and concurrent cropping of buffer regions all sustain some overhead to instantiate the pools for multiprocessing. To minimize total overhead, we have all three of these functions happen within a single multiprocessing pool. This streamlining is possible since each function happens chronologically within the workflow. The optimization provides a scalable performance improvement since higher data decomposition means more multiprocessing instances per function, which, when the functions are streamlined into a single pool, cuts the overhead by a third.

3.4 Optimization Results

We measure the success of our contributions by comparing GEOtiled with and without our optimizations. As seen in Figure 2, we compare runtimes across various tile sizes, where the smaller the tile size, the higher the number of tiles and parallelism used. We make two important observations from our results. First, we see consistently decreasing runtimes as we decrease the tile size and increase parallelism, a scalable behavior. Second, our optimizations run faster for all tile sizes and up to seven times faster for the smallest size.

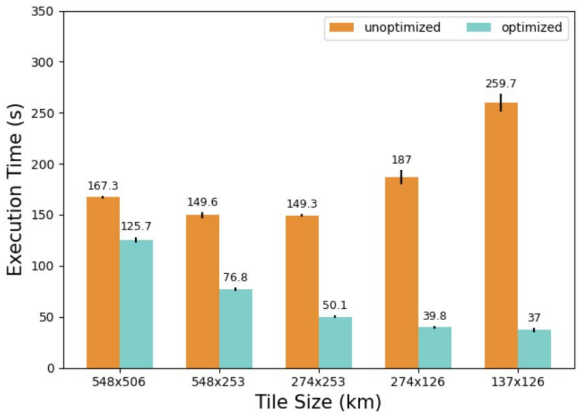


Figure 2: Runtimes of terrain parameter generation using unoptimized vs. optimized GEOtiled across various tile sizes.

4 CONCLUSION

Our results show that GEOtiled 2.0 now produces scalable runtime behavior for higher parallelization and runs up to seven times faster than GEOtiled pre-optimization. Our improvements boost GEOtiled's performance and highlight how powerful data decomposition and parallel computation are for computing terrain parameters. This provides scientists with a software tool for quickly and effectively generating terrain parameters, saving needed time and resources to do their work.

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